

Power meter

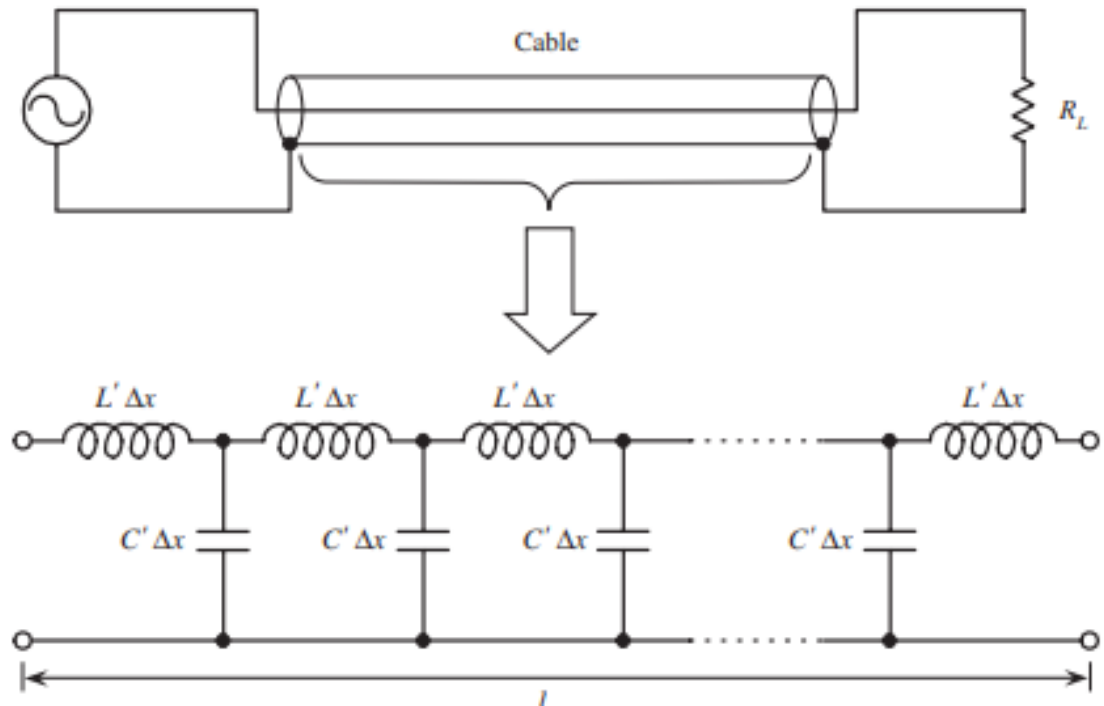
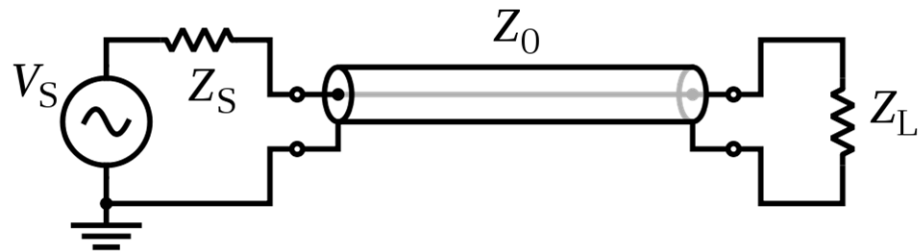
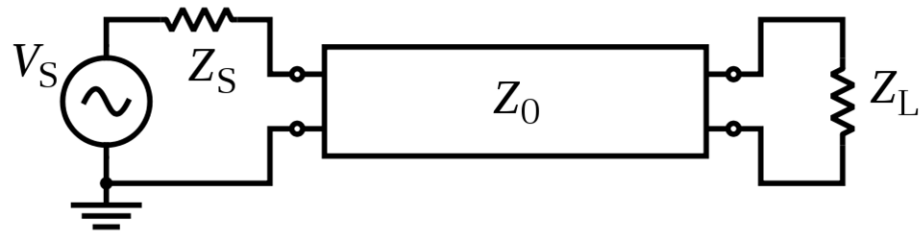
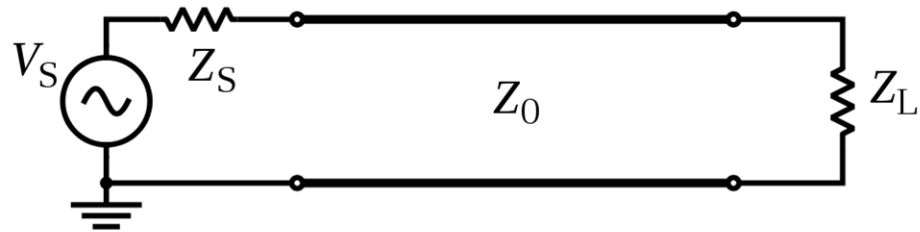
Effektmeter



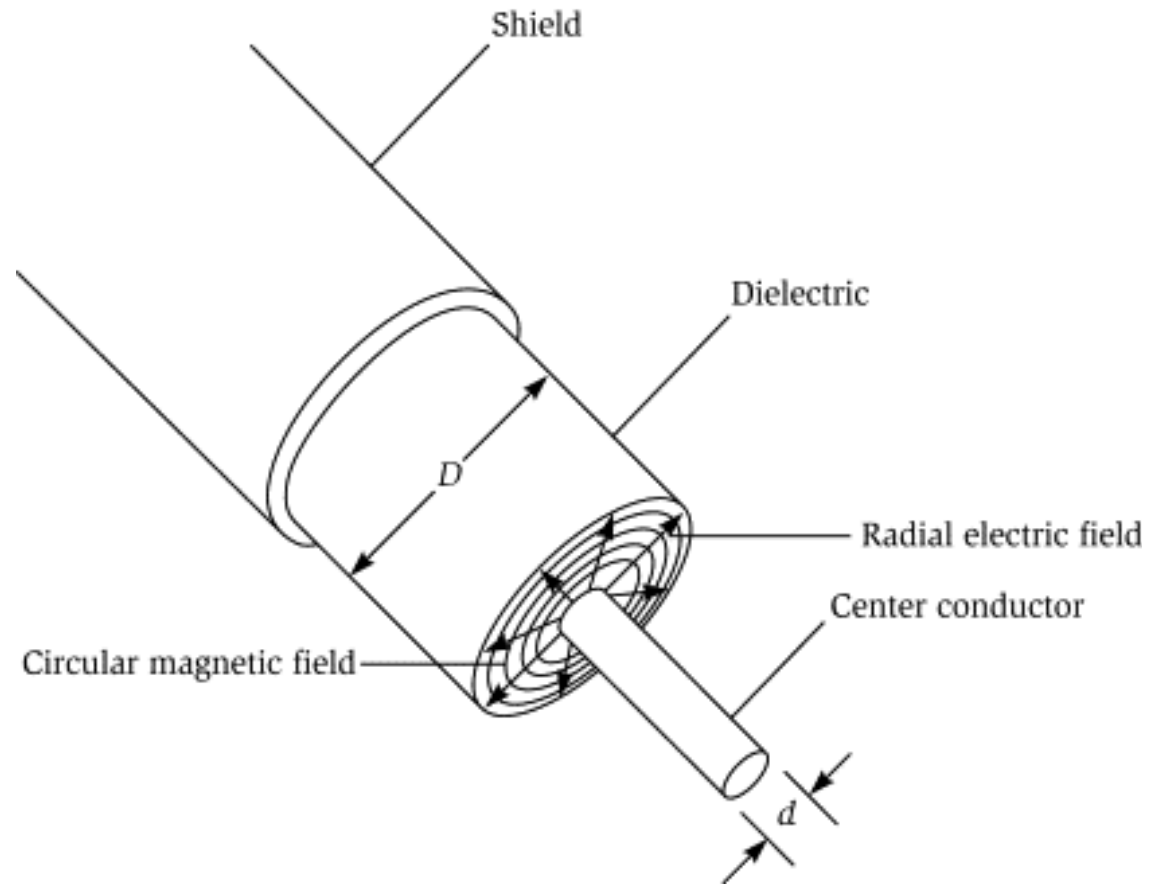
Signalgenerator



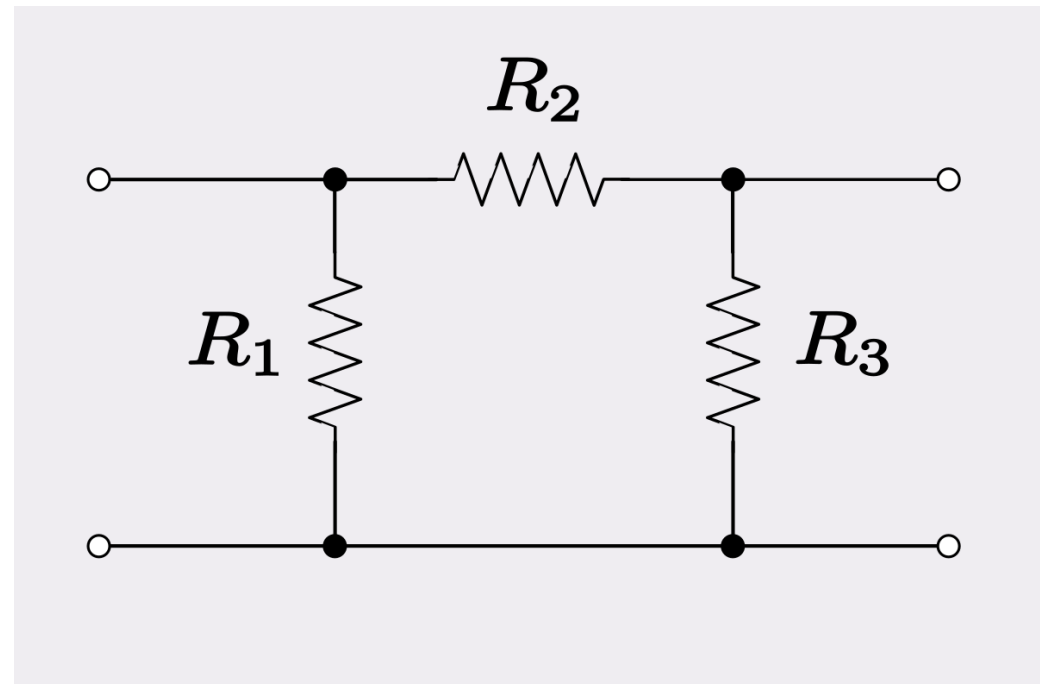
Effekt och karakteristisk impedans



Koaxialkabel



Dämpfsats



Decibel dB

$$U_{dB} = 20 \log_{10}(U_{lin}) \quad \text{Spänning}$$

$$I_{dB} = 20 \log_{10}(I_{lin}) \quad \text{Ström}$$

$$P_{dB} = 10 \log_{10}(P_{lin}) \quad \text{Effekt}$$

$$P = U^2/R = I^2 R \quad \log_{10}(U^2) = \log_{10}(U \cdot U) = \log_{10}(U) + \log_{10}(U) = 2 \log_{10}(U)$$

Decibel dB (snarare så här)

$$U_{dBV} = 20 \log_{10}(U_{lin}/1V)$$

Spänning

$$I_{dBA} = 20 \log_{10}(I_{lin}/1A)$$

Ström

$$P_{dBW} = 10 \log_{10}(P_{lin}/1W)$$

Effekt

$$P_{dBm} = 10 \log_{10}(P_{lin}/1mW)$$

Effekt

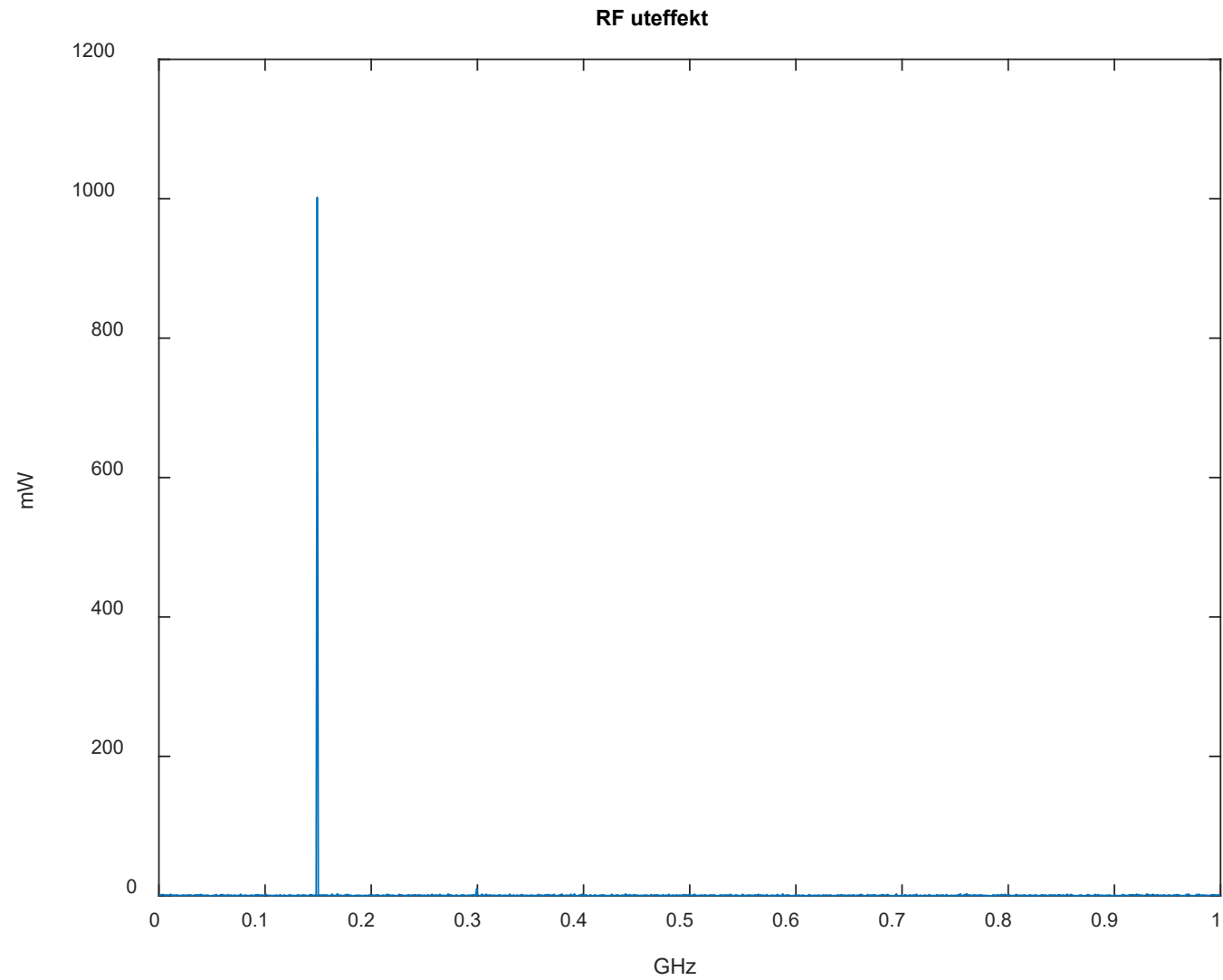
$$A_{dB} = 20 \log_{10}(U_{ut}/U_{in})$$

Spänningsförstärkning

$$A_{dB} = 10 \log_{10}(P_{ut}/P_{in})$$

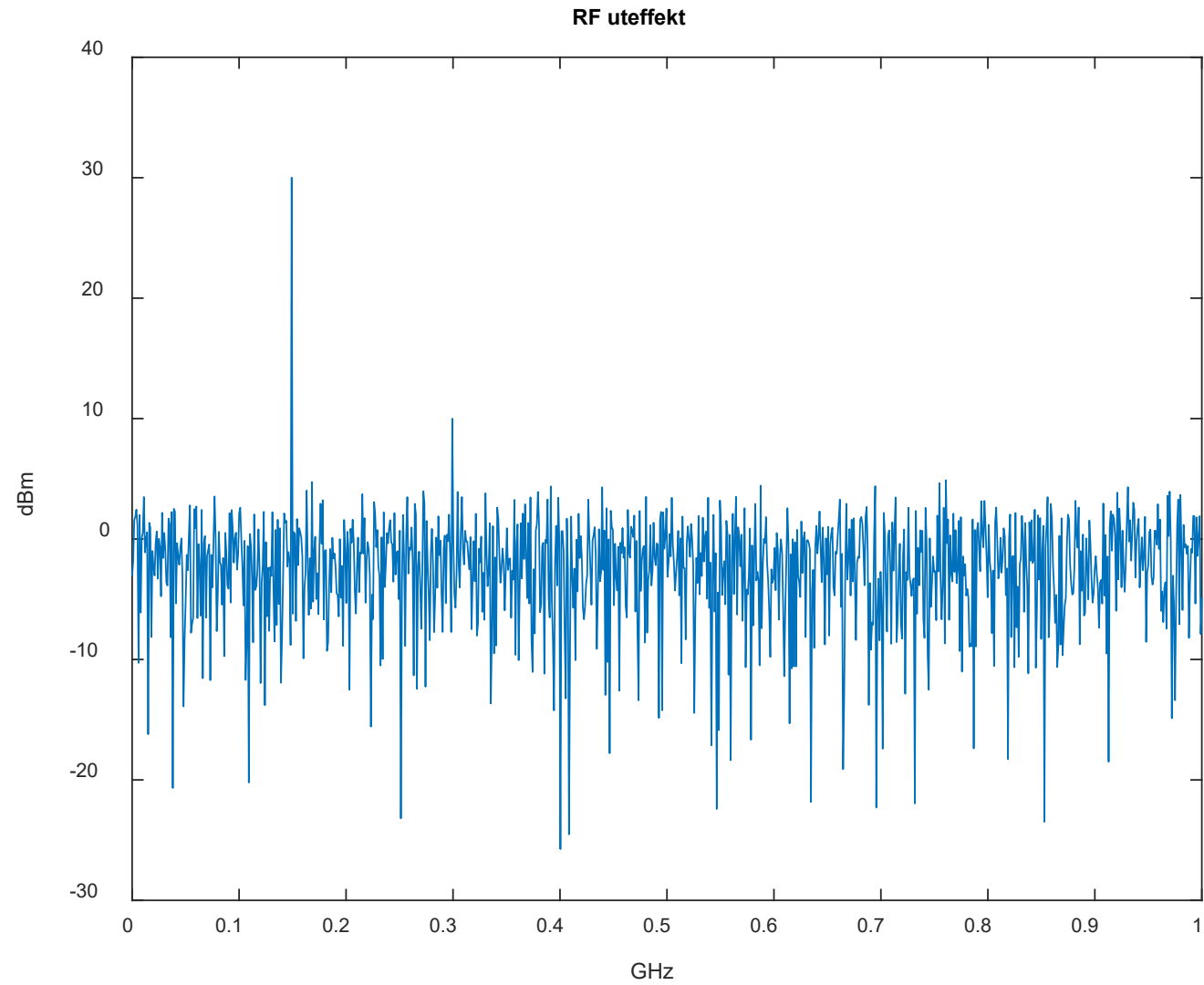
Effektförstärkning

Decibel dB



Decibel

dB



Decibel dB

$$A_{dB} = -X$$

Förstärkning

$$L_{dB} = X$$

Dämpning

$$[dB] + [dB] = [dB]$$

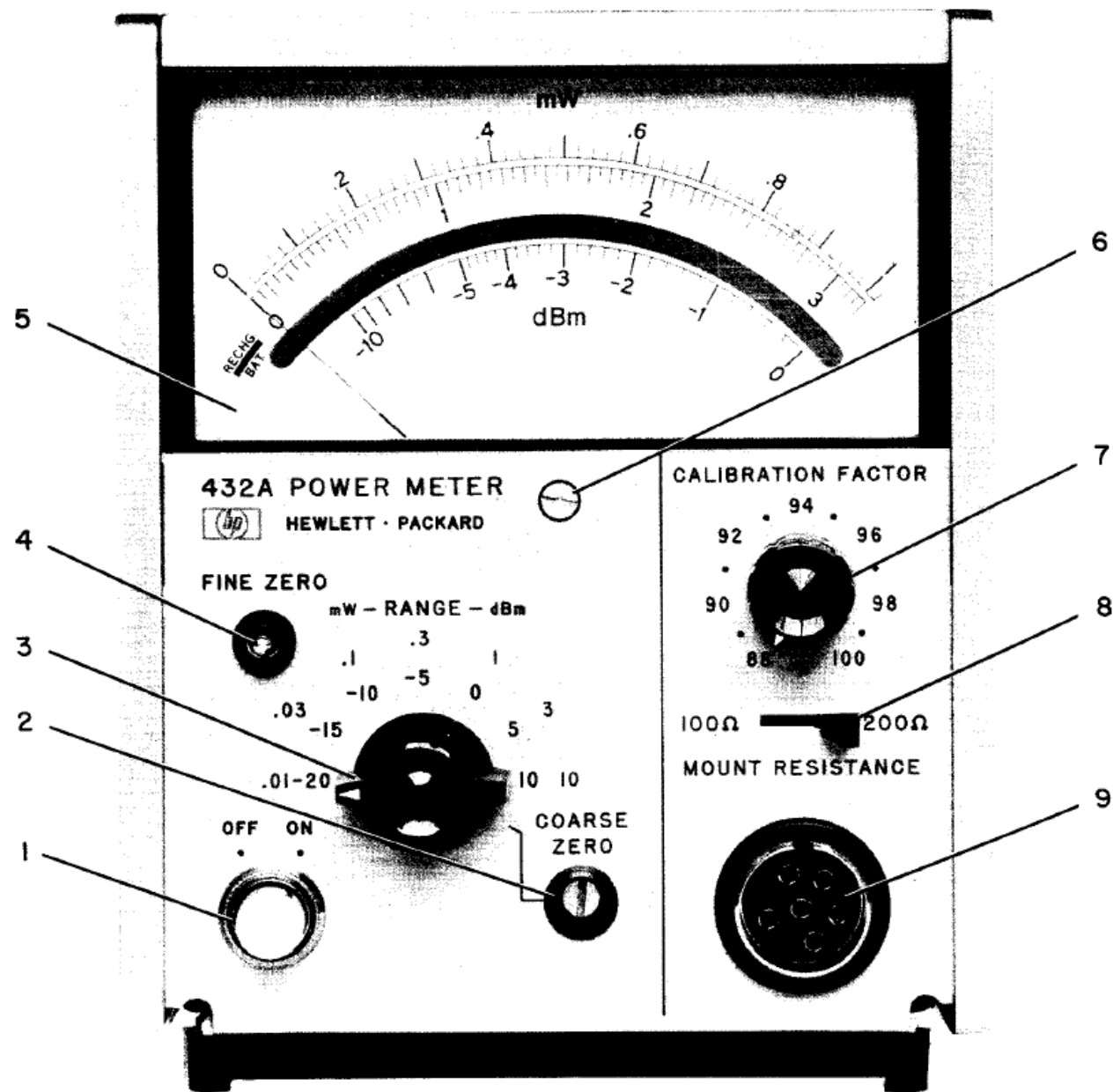
Förstärkning och dämpning

$$[dB] + [dBm] = [dBm]$$

Effekt

$$[dBm] + [dBm] = ?$$

Effekt² , oftast ett misstag av någon



1. **POWER.** Instrument power ON/OFF switch; connects either ac line voltage or internal battery (Option 01 only) to internal voltage regulator circuits. When ac power is on, optional battery charging circuit operates.
2. **COURSE ZERO.** Meter zero adjustment; set the RANGE selector to COURSE ZERO, turn OFF the RF power, and adjust to zero the meter.
3. **RANGE.** Power measurement range selector; selects ranges from 0.01 to 10 milliwatts (-20 to +10 dBm). COURSE ZERO setting is used to zero meter with no power applied to thermistor mount.
4. **FINE ZERO.** Electronic zero that balances the compensation bridge with zero RF input. To zero meter during operation, close the switch momentarily. Be sure that RF power is not applied to the thermistor mount when the FINE ZERO switch is depressed.
5. **Meter.** Indicates power input to thermistor mount in milliwatts and dBm. To use the dBm scale, note the value in dBm of the range in use, and subtract from it the reading on the meter dBm scale.
6. **Mechanical Meter Zero.** Sets meter suspension so that meter indicates zero. To adjust the zero:
 - a. Turn POWER switch off.
 - b. Turn the adjustment screw clockwise until the indicator falls below zero and comes back up to zero again.
 - c. Turn the adjustment very slightly counter-clockwise to free up the mechanism from the adjusting peg.
7. **CALIBRATION FACTOR.** Amplifier gain compensation selector. Set to correspond to the calibration factor printed on the thermistor mount body. See paragraph 3-23 for more information.
8. **MOUNT RESISTANCE.** Selects resistance equal to that of mount in use to balance bridges. Table 1-2 lists Hewlett-Packard thermistor mounts and resistances. Set with meter power OFF, when mount is initially connected to the meter.
9. **Thermistor Mount Cable Connector.** Input connector for 5-1/2 foot cable that connects to the 478A, 8478B, or 486A Thermistor Mounts.

Decibel dB

$$P_{ut_dBm} = P_{in_dBm} - L_{dB}$$

Signal dämpas L dB

$$L_{dB} = P_{in_dBm} - P_{ut_dBm}$$

Beräkna dämpning L (i dB)

Laboration 1

- 1) Beräkna vilken signalamplitud (sinus) som motsvarar en utvecklad effekt av -16dBm i 50 Ohm last.
- 2) Ställ in signalgeneratorns amplitud på det beräknade värdet, och frekvens 2 MHz (sinus).
- 3) Anslut signalgeneratorn till oscilloskopet, med korrekt terminering.
- 4) Avläs amplitud på oscilloskopet och beräkna vilken effekt det motsvarar. Stämmer resultatet överens med 1) ?
- 5) Ställ in effektmeter på rätt mätområde, anslutningsmotstånd (200Ohm), samt kalibreringsfaktor.
- 6) Nolla effektmeter (fine zeroing).
- 7) Koppla loss signalen från oscilloskopet och anslut istället till effektmeter.
- 8) Läs av effektmeter.
- 9) Hur stora relativa effektskillnader är det (dB) mellan signalgeneratorinställning, oscilloskop, och power meter?

